

# SET THEORY

## 1.1 SETS AND SET MEMBERSHIP

The concept of a set is a fundamental one in modern mathematics and is usually a starting point in the development of basic mathematics and its applications in many fields. In the area of management, for example, we often refer to a set of manufactured products, a set of workers in a factory, or a set of alternative decisions open to a manager. A knowledge of set theory is essential for understanding the concepts and techniques that are developed in this book. It will be seen that many complex problems can be solved in a logical and simple way by structuring them in the set theory language. In this chapter we shall discuss the basic notions and operations involved in the theory of sets.

A *set* is a *well defined collection* of objects. Objects forming part of the set are called members or elements of the set. The following are some examples:

### *Example 1.1.1*

- (i) Set of all possible outcomes when we toss a coin.
- (ii) Set of all products that a company manufactures.
- (iii) Set of integers from 1 to 100.
- (iv) Set of all stocks listed in the Bombay Stock Exchange.

A set is represented by listing its elements and enclosing them in braces. For example, the set defined in (i) of Example 1.1.1 is represented by {heads, tails} and the set defined in (iii) by {1, 2, 3, 4, ..., 98, 99, 100}.

We require that the collection of the objects forming part of a set is well defined. This means that we should be able to decide without ambiguity whether or not an element is in a given set. If we define a set  $A$  as the set of all possible outcomes when two dice are thrown, this set is not

well defined. It depends on how one defines an outcome. If an outcome is defined as the sum of the points on the dice, then  $A$  is  $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ .

If an outcome is defined as the points shown by the first die and the second die, the set  $A$  consists of 36 outcomes like:

|          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| $(1, 1)$ | $(1, 2)$ | $(1, 3)$ | $(1, 4)$ | $(1, 5)$ | $(1, 6)$ |
| $(2, 1)$ | $(2, 2)$ | $(2, 3)$ | $(2, 4)$ | $(2, 5)$ | $(2, 6)$ |
| $(3, 1)$ | $(3, 2)$ | $(3, 3)$ | $(3, 4)$ | $(3, 5)$ | $(3, 6)$ |
| $(4, 1)$ | $(4, 2)$ | $(4, 3)$ | $(4, 4)$ | $(4, 5)$ | $(4, 6)$ |
| $(5, 1)$ | $(5, 2)$ | $(5, 3)$ | $(5, 4)$ | $(5, 5)$ | $(5, 6)$ |
| $(6, 1)$ | $(6, 2)$ | $(6, 3)$ | $(6, 4)$ | $(6, 5)$ | $(6, 6)$ |

The order of elements in a set is not important. Thus, both  $\{1, 2, 3\}$  and  $\{3, 2, 1\}$  represent the same set. Sets will be denoted by capital letters, say  $A, B, C \dots$ , and the elements of the set by small letters, say  $x, y, z, \dots$ . If  $x$  is a member of  $A$ , we write  $x \in A$ . If  $y$  is not a member of  $B$ , we write  $y \notin B$ .

The elements of a set may represent quantitative or qualitative characteristics or both. The elements of set  $A = \{1, 2, 3, 4, 5, 6\}$  represent quantitative characteristics, while set  $B = \{\text{Aggressive, Timid, Sporting, Nervous}\}$  has elements which connote behavioural patterns. The collection  $C = \{(\text{Male, 64 inches}), (\text{Female, 60 inches})\}$  is also a set where each element indicates both the sex and the height of a person.

Sets can be described in several ways. We may either list the elements of a set indicating them within brackets or we may state some characteristic that the elements of the set are required to possess. A few examples of sets defined in both these ways are given below.

#### Example 1.1.2

(i) Set  $A = \{2, 4, 6, 8, 10\}$  can also be described as the set of all even positive integers up to and including 10.

(ii) Set of all departments in a given management institute:  
 $\{\text{Accounting, Marketing, Production, Economics, Quantitative Methods, Business Policy, Finance}\}$ .

(iii)  $B = \{\text{Good item, Defective item}\}$ .

When a set is described by defining first a characteristic that the elements of the set possess, it is customary to write the set as

$$\{x \mid x \text{ possesses the characteristic mentioned}\}.$$

This is to be read as "the set of all elements  $x$  such that  $x$  possesses the characteristic mentioned". For example

$$\{x \mid x \text{ is an outcome when a die is tossed}\}$$

yields the set  $\{1, 2, 3, 4, 5, 6\}$ . This way of representing a set is useful when it is not possible to list all the elements of a set. For example, it is not possible to write completely the set of all positive integers. However, this is written as  $\{x \mid x \text{ is a positive integer}\}$ .

The set of those elements that are of interest in a study is called the *universal set* (*fundamental set* or *master set*) and will be denoted by  $S$ . Examples of the universal set are:

If we wish to study some characteristics of female workers in a factory, we may take for the universal set as the totality of female workers employed by the factory; if we study the details of the trips operated by a bus company, the set of all the trips operated by the bus company will constitute the universal set. The universal set therefore provides the basic frame of reference for further study.

In set theory, it has been found useful to define a set called the *null set* or *empty set*. A set that contains no element is called a null set. Though it is difficult to visualize a set that contains no element this concept, like the zero of the number system, is useful while performing certain algebraic operations on the sets discussed later. The null set is usually defined by the Greek letter  $\phi$ . The sets

$$C = \{x \mid x \text{ is an integer greater than six in an outcome when a die is tossed}\}$$

$$D = \{\text{Set of odd integers divisible by 2}\}$$

are examples of null sets. The null set  $\phi$  should not be confused with the set consisting of the single element 0, i.e.  $\{0\}$ . Note that

$$\{x \mid 3^x = 1\} = \{0\}$$

## 1.2 SUBSETS AND SET EQUALITY

If every member of a set  $B$  is also a member of another set  $A$ , then  $B$  is said to be a subset of  $A$  and this is denoted by  $B \subset A$ . By convention, the empty set  $\phi$  is a subset of every other set.

### Example 1.2.1

$$A = \{x \mid x \text{ is a positive integer } \leq 100\}$$

$$B = \{x \mid x \text{ is a positive integer } \leq 50\}$$

Here,  $B \subset A$ . If we define

$$C = \{x \mid x \text{ is a positive integer } \leq 720 \text{ and divisible by 4}\},$$

$C$  is not a subset of  $A$ , since  $180 \in C$  and  $180 \notin A$ . Note that  $A$  is also not a subset of  $C$ , since  $29 \in A$  and  $29 \notin C$ . Thus, neither  $C$  is a subset of  $A$  nor  $A$  is a subset of  $C$ .

## Number of Subsets that can be Formed from a Given Set $A$

Suppose a set  $A = \{1, 2, 3, 4\}$ . How many subsets of  $A$  can be formed? We can enumerate all the subsets as follows:

$\phi$  (Recall that  $\phi$  is a subset of every other set),

$\{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 3, 4\}$

giving, in all, 16 subsets. It will be demonstrated later that if a set  $A$  has  $n$  elements, the total number of subsets is given by the expression  $2^n$ . Even

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for a small value of  $n = 20$ ,  $2^n = 1,048,576$ , which gives a very large number of subsets that can be formed from a set consisting of 20 elements.

**Equal Sets** From the definition of subsets, we can define equality of two sets. *Two sets are said to be equal if each is a subset of the other.* Alternatively, two sets are equal if they contain the same elements. Clearly, two equal sets contain the same number of elements. However, two sets may contain the same number of elements, but they may not be equal sets.